Introduction

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Slides mostly follow "Introduction to Machine Learning, 3rd edition Ethen Alpaydin

Some parts of the material can be found in Machine learning with Scikit-Learn & tensor flow by Aurelien Geron

Machine Learning



- Field of study that gives computers the ability to learn without being explicitly programmed. – ***Arthur Samuel, 1959***

- A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience – ***Tom Mitchell, 1997***

Traditional Approach vs Machine Learning



- Write rules-> evaluate -> analyze -> study-> write rules
- Train ML(data)-> evaluate-> analyze-> study->trainML(Data)
- Examples: Spam filter, Speech to text, text identification etc.
- Humans can learn from Machine Learning.

Machine learning is used when..

- Human expertise does not exist (navigating on Mars)
- Humans are unable to explain their expertise (speech recognition)
- Solution changes in time (routing on a computer network)
- Solution needs to be adapted to particular cases (user biometrics)

What do we mean by learning ?

- Learning general models from a data of particular examples
- Example in retail: Customer transactions to consumer behavior:

  People who bought "Blink" also bought "Outliers" (www.amazon.com)

- Build a model that is *a good and useful approximation* to the data.

Data Mining

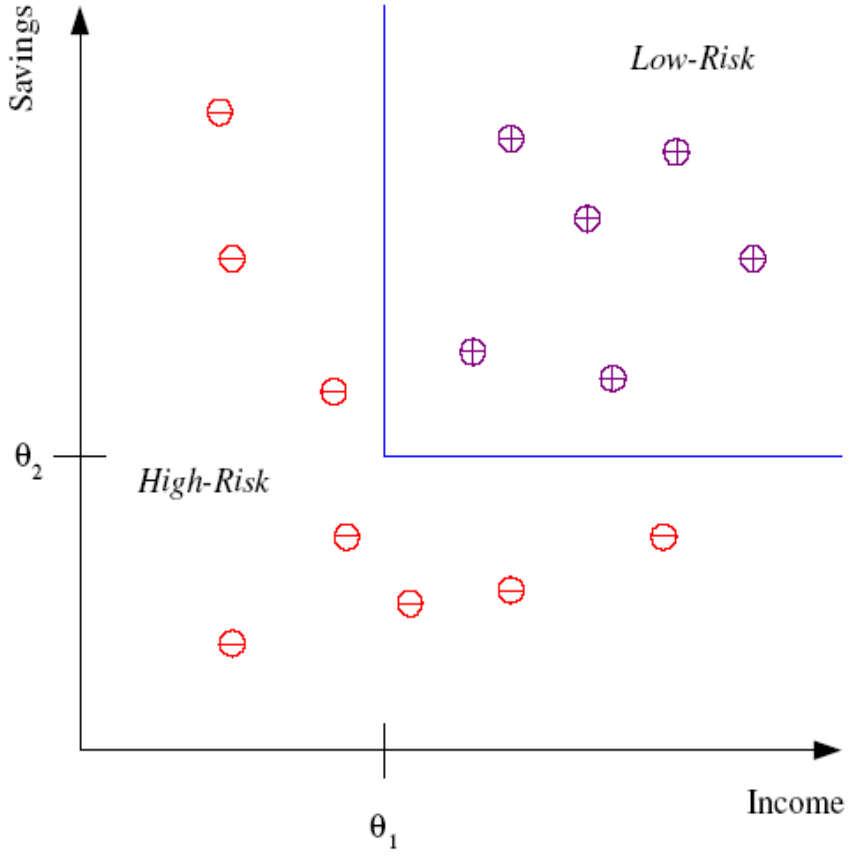
- Retail: Market basket analysis, Customer relationship management (CRM)
- Finance: Credit scoring, fraud detection
- Manufacturing: Control, robotics, troubleshooting
- Medicine: Medical diagnosis
- Telecommunications: Spam filters, intrusion detection
- Web mining: Search engines
- …

Different types of learnings

- Association
- Supervised Learning
  - Classification
  - Regression
- Unsupervised Learning
- Reinforcement Learning

Classification

- Example: Credit scoring
- Differentiating between low-risk and high-risk customers from their *income* and *savings*



Discriminant: IF *income* > θ_1 AND *savings* > θ_2
THEN low-risk ELSE high-risk

Classification: Applications

- Aka Pattern recognition
- Face recognition: Pose, lighting, occlusion (glasses, beard), make-up, hair style
- Character recognition: Different handwriting styles.
- Speech recognition: Temporal dependency.
- Medical diagnosis: From symptoms to illnesses
- Biometrics: Recognition/authentication using physical and/or behavioral characteristics: Face, iris, signature, etc
- …

Regression

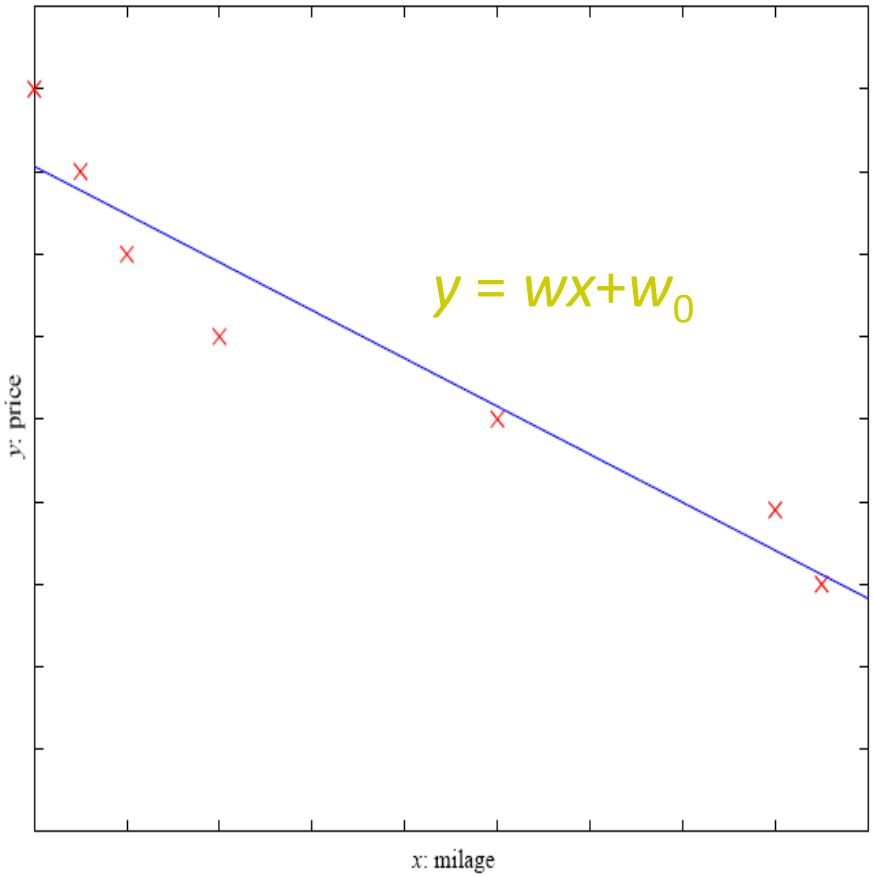
- Example: Price of a used car
- x : car attributes

  y : price

  $y = g(x \mid \theta)$

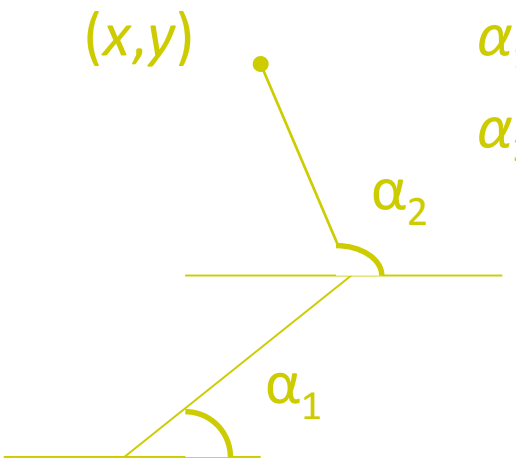
  $g(\ )$ model,

  θ parameters



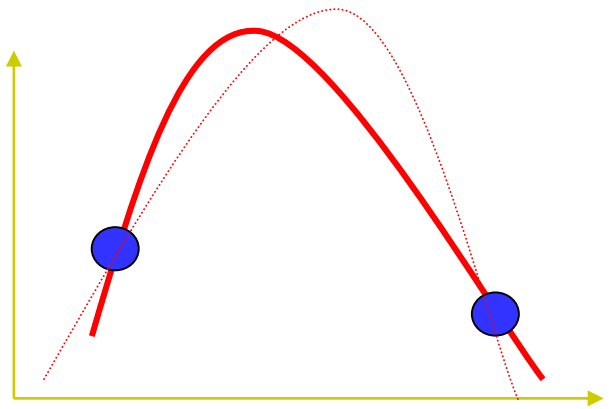
Regression Applications

- Navigating a car: Angle of the steering
- Kinematics of a robot arm



$\alpha_1 = g_1(x,y)$

$\alpha_2 = g_2(x,y)$



- Response surface design



Supervised Learning: Uses

- Prediction of future cases: Use the rule to predict the output for future inputs
- Knowledge extraction: The rule is easy to understand
- Compression: The rule is simpler than the data it explains
- Outlier detection: Exceptions that are not covered by the rule, e.g., fraud

Supervised Learning: Algorithms

- K-nearest neighbors
- Linear regression
- Logistic regression
- Support Vector Machines (SVMs)
- Decision Trees and Random Forests
- Neural Networks

Unsupervised Learning

- Learning “what normally happens”
- No output
- Clustering: Grouping similar instances
- Example applications
  - Segmentation
  - Anomaly detection
  - Dimensionality reduction

Unsupervised Learning: algorithms

- Clustering
  - K-Means
  - Hierarchical Cluster Analysis
  - Expectation maximization
- Visualization and dimensionality reduction
  - Principal component analysis (PCA)
  - Kernel PCA
  - t-distributed stochastic neighbor embedding (t-SNE)
- Association rule learning
  - Apriori
  - Eclat

Semisupervised Learning

- Algorithms for partially labeled data
- Example application: Photo hosting services
- Most algorithms are combination of supervised and unsupervised learning algorithm

Reinforcement Learning

- Learning a policy: A sequence of outputs
- No supervised output but delayed reward
- Systems are called agents
- Observe->select action from policty-> action-> get reward/penalty-> update policy .. repeate
- Example applications:
  - Credit assignment problem
  - Game playing
  - Robot in a maze
  - Multiple agents, partial observability, ...

Other ML system classifications..

- Batch vs online
  - Batch: must use all the available data
  - Online: system learn incrementally by using data instances sequentially
- Instance vs Model based
  - Instance: Learn the examples by heart, then generalize to new cases using similarity measures
  - Model: Use the model to generalize/predict

Resources: Datasets

- UCI Repository: http://www.ics.uci.edu/~mlearn/MLRepository.html
- UCI KDD Archive: http://kdd.ics.uci.edu/summary.data.application.html
- Statlib: http://lib.stat.cmu.edu/
- Delve: http://www.cs.utoronto.ca/~delve/

Resources: Journals

- Journal of Machine Learning Research www.jmlr.org
- Machine Learning
- Neural Computation
- Neural Networks
- IEEE Transactions on Neural Networks
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- Annals of Statistics
- Journal of the American Statistical Association
- ...

Resources: Conferences

- International Conference on Machine Learning (ICML)
- European Conference on Machine Learning (ECML)
- Neural Information Processing Systems (NIPS)
- Uncertainty in Artificial Intelligence (UAI)
- Computational Learning Theory (COLT)
- International Conference on Artificial Neural Networks (ICANN)
- International Conference on AI & Statistics (AISTATS)
- International Conference on Pattern Recognition (ICPR)
- …